Which limits of Newton's construction reach the zero branch

The thesis under test (user, 2026-10-01): Newton's mechanics exists only with a positive action scale, no zero branch is continuously reachable, and the family of theories has a gap in the sense of the mass gap. This note states the thesis as a claim about limits and proves, in the simplest cases, which of Newton's own limits reach the zero theory and which do not. Result: with Gaussian preparations and Gaussian marks, Galileo's comparison has a record family that is the same for every value of the constant (Theorem 1), so there is no gap there. For a determinate preparation, one definite velocity at each place, the complete coordinate record has no limit as the constant tends to zero wherever two or more of its streams cross with distinct actions, or a limit different from Newton's where they cross with equal actions, while every smeared record converges to Newton's branch-weighted density (Theorem 2, any finite partition; Theorem 3, exactly, for inertial motion with a graded velocity inside its fold interval; Theorem 4, for a Kepler swarm whose windings overlap). The same swarm attached as an incoherent mixture of coherent states whose widths vanish with h has a complete record that converges to Newton's branch-weighted density at every fixed time (Theorem 5, exact for quadratic forces), so the gap requires coherence between the launch points of the crossing streams. The gap thesis therefore holds in the topology of complete records of determinate preparations wherever streams cross and the preparation is coherent, and fails in the topology of smeared records, for statistical preparations, and for incoherent attachments. Refinement with records is the one Newtonian limit in which the zero theory is the regular side. Proofs are written derivations by Claude Fable (2026-10-01); the stationary phase, Poisson summation and almost-periodicity facts used are standard and named. Refereed by GPT-6.1 Sol (Codex) on 2026-10-01: Theorem 1, the Lemma, Theorem 2's expansion and smeared limit, Theorem 3(a), the Poisson step and Theorem 4(a) accepted; the frequency condition of Theorem 2(i), the resolution and thickness corollaries, the Airy constant and the Kepler times refined; the universal late-time claims first made in Theorems 3(b) and 4(b) rejected by counterexample and replaced by the local statements now printed. The corrections are applied below. Theorem 5 (§4c) was refereed separately by GPT-6.1 Sol (Codex) in three bounded passes on 2026-10-01: REFINE (critical values, uniformity, part (c), units), REFINE (one modulus), then ACCEPT on the statement and proof as printed; the reports are kept in `reviews/2026-10-01-theorem5-coherence-necessity-referee.md`.

Throughout, h denotes the reduced phase constant, as in the 1998-conjecture note; the recording notes write $\kappa = h/2$.

1. The thesis as a statement about limits

For each $h > 0$ let T_h be the theory with phase constant h: its dynamics and its record law. The dilation $h \mapsto \mu h$ is an isomorphism (fifth-postulate note, Theorem B and §7; Rivero 1998, eq. (10), full read), so $\{T_h : h > 0\}$ is one theory in different units. The zero theory T_0 is Newton's. The thesis says that T_0 is not the limit of T_h as $h \to 0$ in any topology intrinsic to Newton's construction: the scale is generated by the construction, and zero is isolated. This is the mechanical form of a mass gap, where the continuum limit generates the scale and the massless theory is not reached by the coupling.

Known results fix what the thesis can mean. The limit $h \to 0$ is singular, with phases $e^{iS/h}$ that have no pointwise limit [@Berry2002; metadata]. Yet in weak topologies and for finite times the classical theory is reached: coherent states follow classical trajectories for finite times [@Hepp1974; metadata], Egorov's theorem transports smooth observables classically up to $O(h)$ for finite times, and Wigner measures give the weak limits of densities [@LionsPaul1993; metadata]. So the thesis is false for smeared records at finite time, and can be true only in stronger topologies. Newton's construction supplies three candidates: complete records, the densities themselves rather than their smearings; the all-time limit; and refinement, with or without records at the inserted times. The rest of the note tests each.

Deformation families cannot see a gap. The Moyal family, the Fisher field law and its sheet realizations (score-constrained note, §§6–8) are continuous in the constant by construction and contain the zero branch. The tangent groupoid glues the $\varepsilon = 0$ fibre continuously to the $\varepsilon > 0$ fibres (fifth-postulate note, §7), so the kinematics is continuous at zero as well. A gap, if it exists, lives in the dynamics and the records.

2. Theorem 1: Galileo's comparison has no gap

Theorem 1 (Gaussian preparations, quadratic forces, Gaussian marks). Let $H = p^2/2m + V$ with $\deg V \le 2$ (inertia, uniform gravity, Hooke's force). For each admissible $h \ge 0$ prepare the body with the Gaussian phase-space law $N(z_0, \Sigma)$, $\Sigma > 0$ fixed; for $h > 0$ this is the Gaussian state with that Wigner function, which exists iff $\det \Sigma \ge h^2/4$. Mark the body at times t_j by Gaussian coordinate copies of widths σ_j, each read or forgotten; the pointers are fresh, independent, minimal-uncertainty and unchirped, with coordinate variance σ_j^2 and momentum variance $h^2/(4\sigma_j^2)$, coupled by the shear $z \mapsto z + q$, $p \mapsto p - \pi$. Then the joint law of all read records and of the final coordinate is, for every admissible h and every finite list of marks, the classical Gaussian process of T_0 with one addition: each mark, read or not, adds to the body momentum an independent centred Gaussian kick of variance $h^2/(4\sigma_j^2)$. The family is continuous in h and equals T_0 at $h = 0$. (A chirped or non-minimal pointer would give a different channel; the conventions are part of the statement.)

Proof. For $\deg V \le 2$ every Moyal flow is the classical affine symplectic flow, and the Wigner function is transported classically (fifth-postulate note, Theorem A(b)). A coordinate copy is generated by $q\pi$, quadratic in the joint canonical variables, so the same holds for the body–pointer shear. A Gaussian Wigner function of a product state is a positive density, and reading the pointer coordinate is classical conditioning of that density. The pointer's momentum π has variance $h^2/(4\sigma_j^2)$ and is independent of its coordinate for a minimal Gaussian pointer, so the shear transfers to the body a kick of that variance whether or not the coordinate is read; this is Proposition 1 of the parallel-move note for the forgotten case, and the read case adds the classical Gaussian conditioning on $r = q + z$. Composition in time is the classical flow between marks. The variance $h^2/(4\sigma_j^2)$ is continuous in h and zero at $h = 0$. □

So on the comparison as the Planck paper sets it, the record family differs from T_0 only by kicks that vanish continuously with h, for every finite list of marks. The zero branch is reached: there is no gap in Galileo's parabola with statistical preparations. (The theorem is a finite calculation; infinite refinement is the separate result that follows.) The same holds for refinement with records at every inserted time: the fixed-h limit exists iff $\sum_j \sigma_j^{-2} < \infty$ (parallel-move note), the $h = 0$ limit always exists, and the $h \to 0$ limit at any fixed partition is continuous. In this limit the zero theory is the regular side and $h > 0$ the singular one.

3. Theorem 2: determinate preparations with several paths

Take the symmetric discrete action of the 1998-conjecture note, (1) there, on a partition with N cells of lengths τ_j, smooth V, fixed initial point $q_0 = x$, record variable $y = q_N$, and internal vertices $z = (q_1, \dots, q_{N-1}) \in \mathbb{R}^d$, $d = N - 1$. With a cutoff $\chi \in C_c^\infty(\mathbb{R}^d)$ and a smooth amplitude O, the halved functional with the record variable kept free is

$$A_h(y) = (2\pi h)^{-d/2} \int \chi(z) O(z)\, e^{iS_\pi(x,z,y)/h}\, dz. \tag{1}$$

Its modulus squared $|A_h(y)|^2$ is the complete coordinate record at the final time; $\int \varphi(y) |A_h(y)|^2 dy$ with $\varphi \in C_c^\infty$ is a smeared record.

Hypothesis (several nondegenerate paths). On an open interval Y of record values, $S_\pi(x, \cdot, y)$ has exactly $K \ge 2$ critical points $z_a(y)$ in $\operatorname{supp} \chi$, all nondegenerate, with $\chi = 1$ near

each. By the implicit function theorem they depend smoothly on y. Write

$$S_a(y) = S_\pi(x, z_a(y), y), \qquad c_a(y) = \frac{O(z_a(y))\, e^{i\pi\sigma_a/4}}{|\det S''_\pi(z_a(y))|^{1/2}}, \tag{2}$$

with σ_a the Hessian signature.

Lemma (final momenta are distinct). $p_a(y) := S'_a(y)$ equals $m(y - q^{(a)}_{N-1}(y))/\tau_{N-1} - \frac{1}{2}\tau_{N-1}V'(y)$, and $p_a(y) \neq p_b(y)$ for $a \neq b$.

Proof. The envelope formula gives $S'_a(y) = \partial_y S_\pi$ at the critical point, which is the displayed expression. The discrete Euler–Lagrange equation at vertex j, $m(q_{j+1} - q_j)/\tau_j - m(q_j - q_{j-1})/\tau_{j-1} + \frac{1}{2}(\tau_{j-1} + \tau_j)V'(q_j) = 0$, is linear in q_{j-1} with coefficient $m/\tau_{j-1} \neq 0$. So a critical point is determined by (q_{N-1}, q_N) by backward recursion, two distinct critical points have distinct q_{N-1}, and the momenta differ. □

Theorem 2. Under the hypothesis, uniformly for y in compact subsets of Y,

$$|A_h(y)|^2 = \sum_a |c_a(y)|^2 + \sum_{a<b} 2|c_a(y)c_b(y)| \cos\Big(\frac{S_a(y) - S_b(y)}{h} + \theta_{ab}(y)\Big) + O(h). \tag{3}$$

(i) *Complete record.* Fix $y \in Y$ and, for each value ω of $S_a(y) - S_b(y)$ over ordered pairs, put $B_\omega = \sum_{a,b:\, S_a - S_b = \omega} c_a \bar{c}_b$. The limit of $|A_h(y)|^2$ as $h \downarrow 0$ exists iff $B_\omega = 0$ for every $\omega \neq 0$, and then equals $B_0 = \sum_s \big|\sum_{a:\, S_a = s} c_a\big|^2$, summed over the distinct action values. For $K = 2$ with $S_1(y) \neq S_2(y)$ the condition is $c_1(y)c_2(y) = 0$. The limit B_0 is Newton's branch-weighted density $D_{0,\pi}(|O|^2) = \sum_a |O(z_a)|^2/|\det S''_\pi(z_a)|$ only if no two illuminated branches share an action. (ii) *Smeared record.* For every $\varphi \in C_c^\infty(Y)$,

$$\int \varphi(y)|A_h(y)|^2 dy \longrightarrow \sum_a \int \varphi(y) \frac{|O(z_a(y))|^2}{|\det S''_\pi(z_a(y))|} dy \qquad (h \downarrow 0). \tag{4}$$

Proof. Equation (3) is the stationary phase expansion of (1) with nondegenerate critical points, (3) of the 1998-conjecture note, whose remainder is uniform in the parameter y on compact sets where the critical points stay nondegenerate (Hörmander's stationary phase theorem with parameters; standard). Expanding the modulus squared gives (3) with $\theta_{ab} = \pi(\sigma_a - \sigma_b)/4 + \arg O_a - \arg O_b$.

(i) Put $s = 1/h$. The cross sum in (3) is an almost periodic function $f(s)$ in Bohr's sense, a finite trigonometric sum. If $|A_h(y)|^2$ converges as $h \downarrow 0$ then $f(s)$ converges as $s \to \infty$. An almost periodic function with a limit at infinity is constant: for any s_0 and $\delta > 0$ the set of δ-almost periods is relatively dense, so there are $T_n \to \infty$ with $|f(s_0 + T_n) - f(s_0)| < \delta$, and the limit forces $f(s_0) = \lim f$. A constant trigonometric sum has zero coefficient at every nonzero frequency: the Bohr mean of the leading terms of $|A_h|^2$ against $e^{-i\omega s}$ is B_ω, which collects both orientations of each unordered pair, and the $O(h)$ remainder does not affect it. This proves the condition, and the value B_0 of the limit when it exists.

(ii) In the cross term with indices $a \neq b$ the phase $\psi_{ab} = S_a - S_b$ has $\psi'_{ab} = p_a - p_b \neq 0$ on Y by the Lemma. One integration by parts, $\int g\, e^{i\psi/h} dy = ih \int (g/\psi')'\, e^{i\psi/h} dy$ for $g \in C_c^\infty(Y)$, bounds the term by $O(h)$. The diagonal terms and the uniform remainder give (4). □

Corollary (finite resolution). Let $\varphi \geq 0$ be smooth with compact support and $\int \varphi = 1$, and let $\varphi_\delta(y) = \delta^{-1}\varphi((y - y_0)/\delta)$ be a record of resolution δ at $y_0 \in Y$. With $\Delta p > 0$ the minimum of $|p_a - p_b|$ over pairs $a \neq b$ on the support,

$$\int \varphi_\delta |A_h|^2 dy = \sum_a \int \varphi_\delta \frac{|O(z_a)|^2}{|\det S''_\pi(z_a)|} dy + O\Big(\frac{h}{\delta\, \Delta p}\Big) + O(h). \tag{4'}$$

Here the windows stay in a fixed compact subset of Y and the branch data are bounded there; the kernel derivative contributes the $1/\delta$ and the amplitude derivatives the $O(h)$. So a record of resolution δ reaches Newton's density if $h/(\delta\Delta p) \to 0$: fringes of spacing $h/\Delta p$ are then unresolved. The condition is sufficient, not necessary: vanishing or cancelling amplitudes defeat the fringes, and so can a zero of the kernel's Fourier transform at the fringe wavenumber, for instance a symmetric two-bump kernel at $\delta = \pi h/(2r\Delta p)$. The complete-record topology of (i) is the order of limits in which $\delta \to 0$ before $h \to 0$; the smeared topology of (ii) is the opposite order.

Proof. In the integration by parts of (ii), the derivative of φ_δ carries a factor δ^{-1} relative to φ_δ itself, and $\int \varphi_\delta = 1$; the remaining terms are as before. □

Thus the complete record of a determinate preparation with several paths does not reach Newton's density as $h \downarrow 0$, by non-convergence in the generic case of distinct actions and by a wrong limit in the symmetric case of equal actions, while its smearing always does. Theorem 1 of the 1998-conjecture note is the case $d = 1$, $K = 3$, $y = 0$ of (i), including the retained cross term of the two mirror paths; smearing in y removes that term by (ii), because the mirror paths arrive at $y = 0$ with opposite momenta.

3b. The invariant discontinuous at zero: fringe visibility

A mass gap is a number that is positive for every value of the coupling and zero for the free theory. The dilation $h \mapsto \mu h$ forbids any scale from playing that role here, so whatever is discontinuous at zero must be dimensionless. It is the leading contrast of the fringes in (3). This is an interference contrast, not a spectral gap, and it generates no scale; the sense in which it is the thesis's invariant is stated at the end of the section.

Proposition (leading visibility). Let $y_0 \in Y$ be a record point with two illuminated nondegenerate branches, $K = 2$, and write $\rho_a = |c_a(y_0)|^2 = |O(z_a)|^2/|\det S''_\pi(z_a)|$ for Newton's branch densities there, $\rho_1 + \rho_2 > 0$, $\Delta p = |p_1 - p_2|(y_0)$. (i) As $h \downarrow 0$ the complete record near y_0 oscillates with local fringe spacing $2\pi h/\Delta p + O(h^2)$ and visibility

$$V_h = \frac{\max|A_h|^2 - \min|A_h|^2}{\max|A_h|^2 + \min|A_h|^2} = V_0 + O(h), \qquad V_0 = \frac{2\sqrt{\rho_1\rho_2}}{\rho_1 + \rho_2}, \tag{V1}$$

the maximum and minimum taken over one fringe, with the slow variation of the background separated from the interference. The leading term V_0 is fixed by Newton's branch densities and does not depend on h. At $h = 0$ the record is $\rho_1 + \rho_2$ and the visibility is zero. (ii) If $S_1(y_0) \neq S_2(y_0)$, the limit set of $|A_h(y_0)|^2$ as $h \downarrow 0$ is the interval $(\rho_1 + \rho_2)[1 - V_0, 1 + V_0]$. If $S_1(y_0) = S_2(y_0)$ the limit exists and equals $\rho_1 + \rho_2 + 2\sqrt{\rho_1\rho_2}\cos\theta_{12}$. (iii) For a record of resolution δ with kernel φ, the leading cross amplitude is multiplied by $\hat\varphi(\delta\Delta p/h)$, $\hat\varphi(\xi) = \int \varphi(u)e^{i\xi u}du$:

$$V_{\delta,h} = V_0\,|\hat\varphi(\delta\Delta p/h)| + O\big(h + \delta + \delta^2 \sup|\psi''|/h\big), \qquad \psi = S_1 - S_2. \tag{V2}$$

For a Gaussian kernel, with its tails controlled, the factor is $\exp[-(\delta\Delta p/h)^2/2]$: full leading contrast when $\delta\Delta p/h \to 0$, none when $\delta\Delta p/h \to \infty$, partial contrast at finite ratios. Other kernels can have Fourier zeros at finite ratios.

Proof. (i) By (3) with $K = 2$, $|A_h(y)|^2 = \rho_1(y) + \rho_2(y) + 2\sqrt{\rho_1\rho_2}(y)\cos(\psi(y)/h + \theta) + O(h)$ with $\psi' = p_1 - p_2 \neq 0$, uniformly near y_0. Over one fringe the cosine runs through ± 1 while the branch densities change by $O(h)$, which gives the spacing, the extremes $\rho_1 + \rho_2 \pm 2\sqrt{\rho_1\rho_2}$ and (V1). (ii) If $\psi(y_0) \neq 0$ then $\psi(y_0)/h$ is continuous and unbounded in h, so its residue mod 2π takes every value along sequences $h \downarrow 0$ and the cosine takes every value in $[-1, 1]$. If $\psi(y_0) = 0$ the cross term is the constant $2\sqrt{\rho_1\rho_2}\cos\theta_{12}$. (iii) Integrate the cross term against φ_δ with $\psi(y) = \psi(y_0) + \Delta p\,(y - y_0) + O(\delta^2 \sup|\psi''|)$ on the window and the coefficients and phase offset frozen at y_0, which costs $O(\delta)$; the result is $2\sqrt{\rho_1\rho_2}\,\mathrm{Re}[e^{i\psi(y_0)/h + i\theta}\hat\varphi(\delta\Delta p/h)]$ up to the stated error, and the diagonal terms are unchanged. □

With $K \geq 3$ illuminated branches, as inside a fold, each pair carries its own leading contrast $2\sqrt{\rho_a \rho_b}/\sum_c \rho_c$ at its own spacing $2\pi h/|p_a - p_b|$. For inertia with graded velocity (Theorem 3) the branch densities are $a(q_a)^2/|1 + p_0'(q_a)t/m|$; for the Kepler swarm (Theorem 4) they are $a(\phi_a')^2/|1 + I_0'\Omega' t|_a$. In every case V_0 is computed from Newton's multi-stream density and is positive wherever two illuminated streams cross, the same along the whole family $h > 0$ as $h \downarrow 0$, and zero at $h = 0$. In the thesis's sense this is the invariant: the leading contrast is constant along the family and jumps to zero at its endpoint. It is not a spectral gap, it is dimensionless, and it generates no scale; what it measures is the discontinuity of the family at zero in the complete-record topology.

3c. Imprecision actions at both ends

The record is one end of the comparison; the preparation is the other. A determinate swarm attached to the $h > 0$ theory as one state (5) has no momentum thickness; a real swarm has one, and so does the same classical swarm attached as a mixture of localized states, whose thickness is of order $\sqrt{h}$. The following proposition says what a thickness does, in the exact free setting of §4; the general case follows by the same envelope identity.

Proposition (preparation thickness). Shift the initial curve by a momentum η, $\psi_0^\eta = a\, e^{i(S_0 + \eta q)/h}$, and average the record over η with a density $w_\Delta(\eta) = \Delta^{-1} w_0(\eta/\Delta)$ of width Δ, w_0 fixed with controlled tails. Let x be a record point with exactly two illuminated nondegenerate arrivals $q_1 \neq q_2$, uniformly across the averaging range, write $\Delta q_0 = |q_1 - q_2|$ for their launch separation, let ϵ_{amp} bound the variation of the branch coefficients across the range and $B = \sup |\partial_\eta (q_1 - q_2)|$ there. Then the averaged record has leading visibility

$$V_{\Delta,h} = V_0\, |\hat{w}_0(\Delta\, \Delta q_0/h)| + O\big(h + \epsilon_{\text{amp}} + \Delta^2 B/h\big), \qquad \hat{w}_0(\xi) = \int w_0(u) e^{i\xi u} du, \tag{V3}$$

so a Gaussian thickness gives the leading factor $\exp[-(\Delta\, \Delta q_0/h)^2/2]$.

Proof. The arrivals solve $q + (p_0(q) + \eta)t/m = x$ and the stationary actions are $\Phi_a^\eta = m(x - q_a)^2/2t + S_0(q_a) + \eta q_a$. By the envelope identity, $d\Phi_a^\eta/d\eta = q_a(\eta)$, so $\Phi_1^\eta - \Phi_2^\eta = \psi(0) + \eta(q_1 - q_2) + O(\eta^2 B)$. Averaging the cross term of (3) against w_Δ with the coefficients frozen gives $2\sqrt{\rho_1 \rho_2}\, \mathrm{Re}[e^{i\psi(0)/h + i\theta} \hat{w}_\Delta((q_1 - q_2)/h)]$ with $\hat{w}_\Delta(\xi) = \hat{w}_0(\Delta\xi)$, up to the stated errors; the diagonal terms change by $O(\epsilon_{\text{amp}})$. □

With both imprecisions present the leading contrast is $V_0\, |\hat{\varphi}(\delta\Delta p/h)|\, |\hat{w}_0(\Delta\, \Delta q_0/h)|$, provided the joint phase is linearized across both ranges, which adds the condition $\delta\Delta \sup |\partial_x (q_1 - q_2)| \ll h$ to the two curvature conditions. Each argument is an action divided by h: the record imprecision $\delta\, \Delta p$, position resolution times the arrival momentum difference, and the preparation imprecision $\Delta\, \Delta q_0$, momentum thickness times the launch separation. The full leading contrast appears when both imprecision actions are small compared with h; for Gaussian kernels it vanishes when either ratio grows without bound and is partial at finite ratios; other kernels can remove it at finite ratios. This is a different statement from the uncertainty relation, which bounds imprecisions from below by h; here the comparison with Newton needs imprecisions pushed below h. Newton's construction has exact preparations and exact records, both imprecision actions zero, so along the whole family $h > 0$ it sees the leading contrast V_0, and at $h = 0$ none. The two attachments of the same classical swarm are separated by the thickness ratio: the mixture of localized states has $\Delta \sim \sqrt{h}$, so $\Delta\, \Delta q_0/h \to \infty$ and its Gaussian contrast vanishes, while the state (5) has $\Delta = 0$. The state (5) has a definite current velocity p_0/m at each place, not a sharp quantum momentum, and its finite-h Wigner function is not a zero-thickness positive sheet; the premise that selects it over the mixture is determinacy at each h in the sense of a conditional momentum spread of order h (the sheet separation $2\kappa\, \partial_q \log \rho$ of the score note) rather than $\sqrt{h}$. The classical swarm alone does not make that selection.

Proposition (coherence length). Attach the same swarm as a mixture over centres c of components $a_\sigma(q - c) e^{iS_0(q)/h}$ with $a_\sigma(u) = (2\pi\sigma^2)^{-1/4} e^{-u^2/(4\sigma^2)}$, the centres weighted by a

smooth density w on the swarm, with σ independent of h. At a record point with two arrivals $q_1 \neq q_2$, launch separation Δq_0, the leading cross term is the common fringe $2\sqrt{\rho_1\rho_2}\cos(\psi(x)/h + \theta)$ multiplied by

$$\int a_\sigma(q_1 - c)\, a_\sigma(q_2 - c)\, w(c)\, dc = e^{-\Delta q_0^2/(8\sigma^2)}\, \bar{w} + O(\sigma|w'|), \tag{V4}$$

$\bar{w}$ the weight at the midpoint, while the diagonal terms are $\rho_a\, w(q_a) + O(\sigma|w'|)$. So the leading contrast is $V_0\, e^{-\Delta q_0^2/(8\sigma^2)}$ up to the weight ratio: partial, the same for every $h > 0$, and zero at $h = 0$. This is (V3) with the minimal thickness $\Delta = h/(2\sigma)$ of such components, since then $\Delta\, \Delta q_0/h = \Delta q_0/(2\sigma)$ and $\exp[-(\Delta\, \Delta q_0/h)^2/2] = e^{-\Delta q_0^2/(8\sigma^2)}$.

Proof. Every component shares the phase S_0, so its arrivals and stationary actions at x are those of Theorem 3 and its fringe phase $\psi(x)/h$ is common; only the amplitudes $a_\sigma(q_a - c)$ depend on the centre. The components are mixed incoherently, so the record is the w-average of the component records. With $(q_1 - c)^2 + (q_2 - c)^2 = 2(c - m)^2 + \Delta q_0^2/2$, m the midpoint, the Gaussian integral over c gives (V4), and $\int a_\sigma(q_a - c)^2 w(c) dc = w(q_a) + O(\sigma|w'|)$. □

The premise therefore has a sharper form than "one coherent state": an h-independent coherence length σ comparable to the launch separation of the crossing streams. Full contrast as $\sigma/\Delta q_0 \to \infty$, none when $\sigma \to 0$ with h (the $\sqrt{h}$ mixture), and an h-independent partial contrast, hence the same discontinuity at zero, for any fixed $\sigma > 0$. In the optical language this is mutual coherence between the launch points of the two streams. It is this quantity, not purity of the whole swarm, that the second premise must assert at each h.

Proposition (large-action regime). Fix $h > 0$ and the geometry of the comparison, and scale the masses by λ at fixed velocities, fixed record resolution δ and fixed velocity imprecision Δv of the preparation. Then momenta, actions, the arrival momentum difference Δp and the momentum thickness $\Delta = \lambda m \Delta v$ all scale by λ, the arrival points, branch densities and V_0 are unchanged, and the two imprecision actions $\delta\Delta p$ and $\Delta\, \Delta q_0$ grow like λ. For Gaussian kernels the leading contrast is $V_0 \exp[-\lambda^2(\delta\Delta p_1/h)^2/2] \exp[-\lambda^2(m\Delta v\, \Delta q_0/h)^2/2]$, with Δp_1 the difference at $\lambda = 1$, and tends to zero as $\lambda \to \infty$; the record converges to Newton's branch-weighted density by Theorem 2(ii) and (4′). The large-action regime at fixed h is therefore the regime in which Newton's mechanics is recovered, and it is defined by comparison with h: the same family read as $h \to 0$ with exact preparations and records, Theorems 2–4, has no limit.

Proof. Under the scaling the classical trajectories are unchanged, $S \mapsto \lambda S$, and the arguments of $\hat{\varphi}$ and $\hat{w}_0$ in (V2) and (V3) are multiplied by λ; the branch densities are ratios of launch weights to Jacobians of the unchanged arrival map. □

4. Theorem 3: inertial motion with a graded velocity

Newton's simplest motion already shows the gap when the preparation is determinate and its velocity varies along it. Let $h > 0$, $V = 0$, and prepare

$$\psi_0(q) = a(q) e^{iS_0(q)/h}, \qquad a \in C_c^\infty(\mathbb{R}) \text{ real}, \qquad p_0 = S_0'. \tag{5}$$

Each particle at q has the definite velocity $p_0(q)/m$. The free propagator is exact, so

$$\psi_t(x) = \Big(\frac{m}{2\pi i h t}\Big)^{1/2} \int a(q)\, e^{i\Phi_x(q)/h} dq, \qquad \Phi_x(q) = \frac{m(x - q)^2}{2t} + S_0(q). \tag{6}$$

Critical points of Φ_x are the arrivals $q + p_0(q)t/m = x$, and $\Phi_x''(q) = m/t + p_0'(q)$. With $M_- = \inf p_0'$ and $M_+ = \sup p_0'$ over the support of a, define the fold interval by

$$t_* = \frac{m}{\max(0, -M_-)}, \qquad t^* = \frac{m}{\max(0, -M_+)}, \qquad t_* \leq t^* \leq \infty, \tag{7}$$

with $m/0 = \infty$: for $t_* < t < t^*$ the derivative $1 + p_0'(q)t/m$ takes both signs on the support, and outside that interval it does not.

Theorem 3. (a) For $0 < t < t_*$ the arrival map $q \mapsto q + p_0(q)t/m$ is a diffeomorphism of $\mathbb{R}$, every x has one arrival $q_*(x)$, and $|\psi_t(x)|^2 \to a(q_*(x))^2/(1+p_0'(q_*(x))t/m)$ pointwise, the classical transported density, zero outside the transported support: the complete record reaches the zero branch before the fold. The same holds for $t > t^*$ when $t^* < \infty$, with the orientation reversed. (b) For $t_* < t < t^*$ the arrival map is not monotone on the support, and there is an open interval of x whose points have three nondegenerate arrivals $q_1 < q_2 < q_3$ in the support. On any record interval where exactly $K \geq 2$ illuminated nondegenerate arrivals exist, Theorem 2 applies with $d = 1$ and the free coefficients $c_a = \sqrt{m/t}\, e^{-i\pi/4} a(q_a) e^{i\pi\sigma_a/4}/|\Phi_x''(q_a)|^{1/2}$: the complete record converges as $h \downarrow 0$ only under the frequency condition of Theorem 2(i), and every smearing in x converges to the classical multi-branch density $\sum_a a(q_a)^2/|1 + p_0'(q_a)t/m|$. (c) At a fold point $x_c(t)$ with an isolated illuminated cubic critical point, $\Phi_x''(q_c) = 0$, $b = \Phi_x'''(q_c) \neq 0$, $a(q_c) \neq 0$, and no other equally singular contribution, $|\psi_t(x_c)|^2 \sim (2\pi m/t)\,\mathrm{Ai}(0)^2 (2/|b|)^{2/3} a(q_c)^2\, h^{-1/3}$.

Proof. (a) For $t < t_*$, $1 + p_0'(q)t/m \geq 1 - t\max(0, -M_-)/m > 0$, so the arrival map is strictly increasing and onto; one-dimensional stationary phase in (6) gives $|\psi_t|^2 \to (m/t)\, a^2/|\Phi_x''|$, the stated ratio, and zero outside the transported support by nonstationary phase. For $t > t^*$ the derivative is negative throughout and the map is decreasing and onto. (b) For $t_* < t < t^*$ the derivative changes sign on the support, so the map has a local maximum and a local minimum there and, by the intermediate value theorem, the values between them are taken three times; the arrivals are nondegenerate away from the two fold values. The final momenta $\partial_x \Phi_x(q_a) = m(x - q_a)/t = p_0(q_a)$ are distinct because the q_a are distinct and $x = q_a + p_0(q_a)t/m$. Theorem 2 applies with the displayed coefficients, whose moduli squared are the classical branch densities. (c) Stationary phase at a cubic point gives $|\int a\, e^{i\Phi/h} dq| \sim 2\pi a(q_c)\mathrm{Ai}(0)(2h/|b|)^{1/3}$ against the prefactor $(m/2\pi h t)^{1/2}$; squaring gives the constant and the power. □

A swarm with an affine velocity profile, $p_0 = -\beta q$, has $M_- = M_+ = -\beta$, so $t_* = t^* = m/\beta$: it passes through a focal instant and never folds, and its complete record reaches the zero branch at every other time. A swarm with faster particles behind slower ones and a non-affine profile folds for $t_* < t < t^*$, with $t^* = \infty$ when p_0' is somewhere nonnegative. Inside the fold interval the zero branch is unreachable by complete records on the three-stream interval; the fringe spacing $2\pi h/|p_a - p_b|$ tends to zero without the record settling, and the classical multi-stream density is recovered only by smearing, an operation external to the determinate construction. The classical density is singular at the fold point; its spatial divergence and the $h^{-1/3}$ divergence of (c) are different limits.

4b. Theorem 4: central forces on the radial cycle, Hooke against Kepler

Newton's central forces supply the case the thesis was made for. Fix the angular momentum $\ell > 0$. The radial motion is one degree of freedom with action $I = I_r$ and angle ϕ on the circle, Hamiltonian $E(I)$, frequency $\Omega(I) = E'(I)$ and radial period $T = 2\pi/\Omega$. The 1998-conjecture note, (10)–(11) gives, for Hooke's force (Book I, Proposition X), $E = \omega(2I + \ell)$, so $\Omega = 2\omega$ and $\Omega' = 0$, the isochrony; and for Kepler's,

$$E = -\frac{mk^2}{2(I+\ell)^2}, \qquad \Omega = \frac{mk^2}{(I+\ell)^3}, \qquad \Omega' = -\frac{3mk^2}{(I+\ell)^4}, \qquad T = 2\pi\sqrt{\frac{ma^3}{k}}. \tag{8}$$

For Kepler the angle ϕ is the mean anomaly, which Kepler's equation turns into position on the ellipse: it is the record astronomy makes.

Model. Take $H_h = E(-ih\partial_\phi)$ on the circle with the twisted periodicity $\psi(\phi + 2\pi) = e^{2\pi i\alpha}\psi(\phi)$, α the Maslov shift of the radial cycle ($\alpha = \frac{1}{2}$), so that the spectrum is $E(h(n + \alpha))$, the Keller–Maslov values, which for Hooke and Kepler are the exact planar spectra ((9)–(11) of that note). Extend E outside the physical range of I by a real smooth function with polynomially controlled derivatives; for preparations whose actions lie compactly inside the physical range, away from the circular boundary $I = 0$, this changes the evolved state by $O(h^\infty)$. The model stipulates the

angle record; agreement of its spectrum with the planar one does not by itself identify its angle density with a physical radial or astronomical measurement, and the physical subset of modes also carries the angular condition $\ell/h = |n_\theta|$. The determinate preparation is a swarm,

$$\psi_0(\phi) = a(\phi)e^{iS_0(\phi)/h}, \qquad a \in C_c^\infty((-\pi,\pi)), \qquad I_0 = S_0', \tag{9}$$

each body at ϕ on the orbit of action $I_0(\phi) > 0$, with $I_0(\operatorname{supp} a)$ compact inside the physical range, action spread $\Delta I = \max I_0 - \min I_0$ on the support and support length $|a| < 2\pi$. Poisson summation over the eigenvalue index gives exactly

$$\psi_t(\phi) = \sum_{k\in\mathbb{Z}} \frac{e^{-2\pi ik\alpha}}{2\pi h} \iint a(\phi')\, e^{i\Phi_k(\phi',I)/h}\, d\phi'\, dI, \qquad \Phi_k = S_0(\phi') + I(\phi - \phi' + 2\pi k) - E(I)t. \tag{10}$$

The critical points of Φ_k are $I = I_0(\phi')$ and $\phi - \phi' + 2\pi k = \Omega(I)t$: the body prepared at ϕ' arrives at ϕ after k turns. The Hessian determinant is $-(1 + I_0'(\phi')\Omega'(I_0(\phi'))t)$. Define the fold time and, for I_0 monotone on the support so that $\Omega(I_0)$ is monotone there, the lapping time

$$t_* = \frac{1}{\max(0,\, -I_0'\,\Omega'(I_0))} \in (0,\infty], \qquad t_{\mathrm{lap}} = \frac{2\pi - |a|}{|\Omega(I_0(b)) - \Omega(I_0(a_-))|} \in (0,\infty], \tag{11}$$

with $a_- < b$ the endpoints of the support and $1/0 = \infty$. The image of the support under the arrival map has length at most $|a| + t\,|\Omega(I_0(b)) - \Omega(I_0(a_-))|$, with equality while the map is monotone.

Theorem 4. (a) *Hooke.* E is affine in I, the I-integral in (10) is a delta function, and for every $h > 0$ and every t

$$|\psi_t(\phi)|^2 = a(\phi - 2\omega t)^2 \pmod{2\pi}: \tag{12}$$

the swarm rotates rigidly, the complete anomaly record is the same for every h, and the zero branch is reached at all times. (b) *Kepler, or any* $\Omega' \neq 0$. For $t < \min(t_*, t_{\mathrm{lap}})$ every ϕ has at most one arrival and $|\psi_t(\phi)|^2 \to a(\phi')^2/|1 + I_0'(\phi')\Omega'(I_0(\phi'))t|$ pointwise. If I_0 is monotone on the support and $t_{\mathrm{lap}} < t_*$, which is the condition $(2\pi - |a|)\max I_0' < \Delta I$ when $\Omega' < 0$ and $I_0' > 0$, then for $t_{\mathrm{lap}} < t < t_*$ the arrival map is monotone with image longer than 2π, and every ϕ in the overlap has exactly two arrivals of different winding; distinct arrivals at the same ϕ always have distinct actions I_a, so Theorem 2 and its Corollary apply with ϕ as record variable and I as momentum: the complete anomaly record converges as $h \downarrow 0$ only under the frequency condition of Theorem 2(i), every smeared record converges to the classical density $\sum_a a(\phi_a')^2/|1 + I_0'\Omega' t|_a$, and a record of angular resolution δ reaches it if $h/(\delta \min|I_a - I_b|) \to 0$. The same conclusions hold on any record interval with finitely many illuminated nondegenerate arrivals, however they arise, by folding as in Theorem 3(b) or by overlap of windings; no universal statement is made for later times, since an inverted monotone map can again be injective with a short image. (c) *Kepler times.* Exactly, $t_* = [3mk^2 \max_{\operatorname{supp} a}(I_0'/(I_0 + \ell)^4)]^{-1}$, infinite when the maximum is nonpositive, and for monotone I_0, $t_{\mathrm{lap}} = (2\pi - |a|)/|\Omega(I_0(b)) - \Omega(I_0(a_-))|$. For a narrow swarm, $\Delta I \ll I + \ell$, with a short support, $|a| \ll 2\pi$,

$$t_{\mathrm{lap}} \simeq \frac{2\pi(I+\ell)^4}{3mk^2\,\Delta I} = \frac{T\,(I+\ell)}{3\,\Delta I} = \frac{T^2}{\Delta T}, \qquad t_* \simeq \frac{(I+\ell)^4}{3mk^2 \max I_0'} \quad (\max I_0' > 0), \tag{13}$$

with $\Delta T = 3T\Delta I/(I + \ell)$ the spread of periods: $T^2/\Delta T$ is the leading full-turn dephasing time of such a swarm. For a uniform gradient, $t_{\mathrm{lap}} < t_*$ needs the support to span more than half a turn, $|a| > \pi$; otherwise the fold comes first and the mechanism of Theorem 3(b) applies on the circle. Both times are classical, independent of h.

Proof. The eigenfunctions are $e^{i(n+\alpha)\phi}$ with eigenvalues $E(h(n+\alpha))$, and the preparation's coefficients are $c_n = (2\pi)^{-1}\int a\, e^{i[S_0(\phi') - h(n+\alpha)\phi']/h} d\phi'$. With $\nu = n + \alpha$ and $I = h\nu$ the evolved state is $\sum_n G(n+\alpha)$ for $G(\nu) = (2\pi)^{-1}\int a(\phi') e^{i[S_0(\phi') + h\nu(\phi - \phi') - E(h\nu)t]/h} d\phi'$, a Schwartz function

of ν: for $h\nu$ outside the range of I_0 the ϕ'-phase has no stationary point and integration by parts gives rapid decay, and derivatives in ν only add polynomial factors. Poisson summation, $\sum_n G(n+\alpha) = \sum_k e^{2\pi i k\alpha} \int G(\nu) e^{-2\pi i k\nu} d\nu$, with $k \to -k$ and $d\nu = dI/h$, is (10). (a) For $E = \omega(2I+\ell)$ the I-integral is $(2\pi h)^{-1} \int e^{iI(\phi-\phi'+2\pi k-2\omega t)/h} dI = \delta(\phi - \phi' + 2\pi k - 2\omega t)$, so $\psi_t(\phi) = \sum_k e^{-2\pi i k\alpha - i\omega\ell t/h} a(\phi + 2\pi k - 2\omega t) e^{iS_0(\phi+2\pi k-2\omega t)/h}$; one k contributes at each ϕ because $|a| < 2\pi$, and (12) follows. (b) For $t < t_*$ the arrival map $\phi' \mapsto \phi' + \Omega(I_0(\phi'))t$ has derivative $1 + I_0'\Omega' t > 0$, so it is injective on the support; its image is an interval of length at most $|a| + t|\Omega(I_0(b)) - \Omega(I_0(a_-))|$, which is less than 2π for $t < t_{\rm lap}$, so no two windings overlap: one nondegenerate critical point at each ϕ of the image, none elsewhere up to $O(h^\infty)$. Two-dimensional stationary phase in (10) with the prefactor $(2\pi h)^{-1}$ gives the pointwise limit; the Hessian is $\begin{pmatrix} I_0' & -1 \\ -1 & -t\Omega' \end{pmatrix}$ with determinant $-(1 + tI_0'\Omega')$. For monotone I_0 and $t_{\rm lap} < t < t_*$ the map is still monotone, its image is an interval of length exactly $|a| + t|\Omega(I_0(b)) - \Omega(I_0(a_-))| > 2\pi$, and the points of the circle covered twice are covered by two support points whose windings differ by one. Two arrivals (ϕ_1', k_1), (ϕ_2', k_2) at the same ϕ with the same action I would satisfy $\phi_1' - \phi_2' = 2\pi(k_1 - k_2)$, hence coincide because $|a| < 2\pi$; so distinct arrivals have distinct actions, and $\partial_\phi \Phi_k = I$ at the critical point is the record momentum of Theorem 2. (c) Insert (8) into (11): $|\Omega'|\Delta I$ is the frequency spread for monotone I_0, $T = 2\pi(I+\ell)^3/(mk^2)$, $T \propto (I+\ell)^3$ gives $\Delta T/T = 3\Delta I/(I+\ell)$, and $t_{\rm lap} < t_*$ is (11) rearranged. □

Proposition X and Kepler's law thus fall on opposite sides of the thesis in the record Newton's astronomy actually makes. The isochronous force keeps the zero branch for all time; the Kepler force loses it for a swarm with a spread of periods once its windings overlap, which for a narrow swarm spanning more than half a turn happens at the lapping time, of order $T^2/\Delta T$, the time after which the fast bodies have gained a full turn on the slow ones, and otherwise once the swarm folds. While streams overlap the complete record of anomalies carries fringes of angular spacing $2\pi h/|I_a - I_b|$ that never settle; no claim is made for all later times. A stationary preparation, the whole ring at one action, is an eigenstate and reaches the zero branch trivially; the gap needs a localized swarm with a spread of actions. In the radial record r, by contrast, even Hooke's swarm has two branches with opposite radial momenta whenever it straddles a turning point, so there the gap appears for both forces during each turning passage; the anomaly record is the one that separates them. In the laboratory coordinate x of a one-dimensional Hooke swarm the same holds with the exact harmonic propagator: the arrivals solve $x = q\cos\omega t + p_0(q)\sin\omega t/(m\omega)$ and the fold condition is $\cos\omega t + p_0'(q)\sin\omega t/(m\omega) = 0$, which every non-affine p_0 meets within a half period; an affine p_0 meets it only at a focal instant, where all bodies pass one point and no second branch forms.

One more scope remark. A single body prepared as one packet of angular width σ_ϕ has action spread $\Delta I \sim h/(2\sigma_\phi)$, so its lapping time (11) grows like $1/h$, as does the fold time of a single packet in Theorem 3 whose velocity gradient is set by its own width. At any fixed time, then, a single body's complete record reaches the zero branch; this is the familiar power-law Ehrenfest time of an integrable system. The thesis concerns swarms of h-independent extent, which is what the complete record of an ensemble describes, and Newton's single trajectory is immune at fixed time.

4c. Theorem 5: the incoherent attachment reaches zero

The coherence-length proposition of §3c says what a fixed coherence length does. This section removes the premise entirely and proves that without it the gap is absent: a determinate swarm attached to the $h > 0$ theory as an incoherent mixture of coherent states whose widths vanish with h has a complete coordinate record that converges to Newton's branch-weighted density at every fixed time. The theorem is exact for quadratic forces, the setting of Theorem 3, where the coherent attachment (5) has no limit inside its fold interval. The gap of this note therefore requires the coherence premise.

Setting. Let $H = p^2/2m + V$ with $\deg V \le 2$, so the classical flow Φ_t is affine with linear part $M_t = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix}$ independent of the phase point. The swarm is the curve $\Lambda = \{(c, p_0(c))\}$, $p_0 = S_0'$ smooth, carrying the launch density $w \in C_c(\mathbb{R})$, $w \ge 0$, $\int w = 1$; its classical measure is $\mu_\Lambda = w(c)\,\delta(p - p_0(c))\,dc\,dp$. Its arrival map at time t is

$$\begin{aligned} &F_t(c) = \pi_x \Phi_t(c, p_0(c)), \\ &F_t(c) = c + \frac{p_0(c)\,t}{m} \ \text{(inertia)}, \qquad F_t(c) = c\cos\omega t + \frac{p_0(c)\sin\omega t}{m\omega} \ \text{(Hooke)}. \end{aligned} \tag{14}$$

Newton's record of the swarm is the pushforward $(F_t)_*(w\,dc)$: at a regular value x of F_t, with preimages q_a in the support and $F_t'(q_a) \neq 0$, it has the density $\sum_a w(q_a)/|F_t'(q_a)|$; at an isolated nondegenerate fold with positive launch weight it has a one-sided inverse-square-root singularity, and at other critical values its behaviour depends on the degeneracy and on the weight.

For $h > 0$ attach the swarm as the mixture

$$\varrho_h = \int w(c)\,|\varphi_c^h\rangle\langle\varphi_c^h|\,dc, \qquad \varphi_c^h(q) = (2\pi\sigma_h^2)^{-1/4} \exp\Big[-\frac{(q-c)^2}{4\sigma_h^2} + \frac{i\,p_0(c)(q-c)}{h}\Big], \tag{15}$$

coherent states in the sense of minimum-uncertainty Gaussian packets, squeezed relative to any fixed oscillator, centred at the swarm's phase points with position width σ_h and momentum width $h/(2\sigma_h)$, both vanishing with h: $\sigma_h \to 0$ and $h/\sigma_h \to 0$, for instance $\sigma_h^2 = hL_*/P_*$ with a fixed length L_* and a fixed momentum P_*. The Wigner function of φ_c^h is the Gaussian with mean $(c, p_0(c))$ and covariance $\Sigma_h = \mathrm{diag}(\sigma_h^2,\, h^2/4\sigma_h^2)$; the Wigner function of ϱ_h is its w-average, a positive density that converges weakly to μ_Λ. So (15) is the same determinate swarm, attached without coherence between centres.

Theorem 5 (coherence necessity). Under the setting above, at every fixed t put $s_t(h)^2 = M_{11}^2\sigma_h^2 + M_{12}^2 h^2/(4\sigma_h^2)$; then $s_t(h) > 0$, since the first row of the symplectic M_t cannot vanish, and $s_t(h) \to 0$ as $h \to 0$ at each fixed t. Then: (a) the complete coordinate record of (15) is exactly

$$\rho_h(x,t) = \int w(c)\,N\big(x;\, F_t(c),\, s_t(h)^2\big)\,dc, \tag{16}$$

a positive density with no oscillatory term; (b) $\rho_h(\cdot,t) \to (F_t)_*(w\,dc)$ weakly as $h \to 0$, and at every regular value x of F_t, with w continuous,

$$\rho_h(x,t) \longrightarrow \sum_a \frac{w(q_a)}{|F_t'(q_a)|} \qquad (h \to 0), \tag{17}$$

Newton's branch-weighted density, the convergence being uniform on compact sets of regular values; at a critical value the weak convergence still holds, while the pointwise behaviour depends on the degeneracy and the launch weight: at an isolated nondegenerate fold $F_t''(q_c) \neq 0$ with $w(q_c) > 0$ the local contribution to ρ_h scales as a positive constant times $s_t(h)^{-1/2}$ and the classical density has a one-sided $|x - x_c|^{-1/2}$ singularity, whereas a weight vanishing at the fold can make both finite; (c) in the setting of Theorem 3, inertia with $w = a^2$, $\int a^2 = 1$, on any regular record interval inside the fold interval carrying exactly $K \ge 2$ illuminated arrivals, the incoherent attachment converges to $\sum_a a(q_a)^2/|1 + p_0'(q_a)t/m|$, where the coherent attachment (5) converges only under the frequency condition of Theorem 2(i), with the equal-action value B_0 when it does. The two attachments are different states at finite h, with the same classical limit on smeared records by Theorem 2(ii); they differ in the complete record. Their difference in coherence is exact: the initial kernel of (15) obeys $|\varrho_h(q,q')| \le \|w\|_\infty e^{-(q-q')^2/(8\sigma_h^2)}$, so the coherence between any two fixed distinct launch points vanishes as $\sigma_h \to 0$, while the state (5) has $|\psi_0(q)\psi_0(q')| = |a(q)a(q')|$. The gap of Theorems 2–4 therefore requires coherence between the launch points of the crossing streams.

Proof. (a) For real quadratic V the Moyal corrections to the Wigner equation vanish identically, so every Wigner function obeys $\partial_t W + (p/m)\partial_q W - V'(q)\partial_p W = 0$ and is transported by the classical affine flow, $W_t = W_0 \circ \Phi_{-t}$; this is the content of Theorem A(b) of the fifth-postulate note read on states. Hence the Wigner function of $U_h(t)\varphi_c^h$ is the Gaussian with mean $\Phi_t(c, p_0(c))$ and covariance $M_t \Sigma_h M_t^{\mathsf{T}}$, whose position marginal is the Gaussian $N(x; F_t(c), s_t(h)^2)$ with $s_t(h)^2 = (M_t \Sigma_h M_t^{\mathsf{T}})_{11}$, the displayed expression. Densities of a mixture add with the mixture's weights, which gives (16). Since M_t does not depend on the phase point, $s_t(h)$ does not depend on c, and $s_t(h) \to 0$ because $\sigma_h \to 0$ and $h/\sigma_h \to 0$. (b) *Weak convergence.* For bounded continuous g, $\int g \rho_h \, dx = \int w(c) \big[\int g(x) N(x; F_t(c), s_t^2) dx \big] dc$; the inner integral tends to $g(F_t(c))$ for every c and is bounded by $\sup |g|$, so dominated convergence gives $\int w(c) g(F_t(c)) dc = \int g \, d(F_t)_*(w \, dc)$. *Pointwise convergence.* Let x be a regular value and $q_1, \dots, q_K$ its preimages in $\operatorname{supp} w$, finitely many because they are isolated in a compact set. Choose disjoint neighbourhoods U_a of the q_a on which F_t is a diffeomorphism onto a neighbourhood of x, and $\eta > 0$ with $|F_t(c) - x| \geq \eta$ for $c \in \operatorname{supp} w \setminus \bigcup_a U_a$, which exists because that set is compact and contains no preimage. The contribution of that set to (16) is at most $\|w\|_\infty |\operatorname{supp} w| \, (2\pi s_t^2)^{-1/2} e^{-\eta^2/2 s_t^2} \to 0$. On U_a substitute $u = F_t(c)$: $\int_{U_a} w(c) N(x; F_t(c), s_t^2) dc = \int w(F_t^{-1}(u)) \, N(x; u, s_t^2) \, \frac{du}{|F_t'(F_t^{-1}(u))|} \to \frac{w(q_a)}{|F_t'(q_a)|}$ by continuity of the integrand at $u = x$. Summing over a gives (17). For uniformity on a compact set R of regular values, cover the compact set $F_t^{-1}(R) \cap \operatorname{supp} w$ by finitely many inverse charts of F_t, on which the local densities $w \circ F_t^{-1} / |F_t' \circ F_t^{-1}|$ are uniformly continuous, and choose one $\eta > 0$ separating R from the images of the remaining compact part of the support, which contains no preimage of R; the tail bound is then uniform and the approximate identity converges uniformly on R. At a critical value the substitution fails; the weak convergence of (b) is unaffected. At an isolated nondegenerate fold with $w(q_c) > 0$, $F_t(c) - x_c \simeq \frac{1}{2} F_t''(q_c)(c - q_c)^2$, and the substitution $c - q_c = s_t^{1/2} u$ in (16) gives a local contribution $w(q_c) \, s_t^{-1/2} \int (2\pi)^{-1/2} e^{-F_t''(q_c)^2 u^4/8} du + o(s_t^{-1/2})$, while the classical density near x_c is $w(q_c) |2/(F_t''(q_c)(x - x_c))|^{1/2}$ on the covered side. If instead w vanishes at the fold, both can stay finite: for free motion with $p_0(c) = (c^2 - c)m/t$, so that $F_t(c) = c^2$, and $w(c) = C c^2 \chi(c)$ with a bump χ, the classical density near $x_c = 0$ is $C\sqrt{x}$ and $\rho_h(0, t) = O(s_t^{1/2}) \to 0$. (c) For inertia $M_t = \begin{pmatrix} 1 & t/m \\ 0 & 1 \end{pmatrix}$, so $s_t(h)^2 = \sigma_h^2 + t^2 h^2/(4m^2\sigma_h^2) \to 0$ and $F_t'(c) = 1 + p_0'(c) t/m$; on a regular interval with K illuminated arrivals, (17) is the stated K-branch density. Theorem 3(b) and Theorem 2(i) give the coherent attachment's behaviour and Theorem 2(ii) the common smeared limit. For the kernel bound, (15) gives $|\varphi_c^h(q) \varphi_c^h(q')| = (2\pi\sigma_h^2)^{-1/2} e^{-[(q-c)^2 + (q'-c)^2]/4\sigma_h^2}$, and $(q - c)^2 + (q' - c)^2 = 2(c - \bar{q})^2 + \frac{1}{2}(q - q')^2$ with $\bar{q}$ the midpoint; integrating the Gaussian in c against $w \leq \|w\|_\infty$ gives the bound. □

Scope. The theorem is exact for quadratic forces, which include the free fold of Theorem 3 and Hooke's laboratory record. For a general smooth force the same conclusion, in the weak form of (b), follows at fixed time from the semiclassical propagation of coherent states along classical trajectories [@Hepp1974; metadata] or from the transport of Wigner measures [@LionsPaul1993; metadata], under the hypotheses those theorems place on the potential, the flow and the admissible squeezing of σ_h; Kepler's $1/r$ lies outside the usual symbol classes, and the angle-record model of Theorem 4 with a polynomially controlled E lies inside them. Those extensions are cited, not proved here. What is proved is the statement the thesis needs: in the very setting where the coherent attachment shows the gap, the incoherent attachment of the same determinate swarm does not.

5. All-time invariants

Two quantities already in the corpus show the other non-commuting pair of limits, $h \to 0$ against $t \to \infty$. For the free Gaussian, the expected number of sign reversals of the two-sheet realization over all time is $\pi\varphi(1)$ for every $h > 0$ and zero at $h = 0$ (recording note, (150)); the total phase

arctan u of (143) there reaches $\pi/2$ for every $h > 0$ and is zero at $h = 0$. Both are limits taken at infinite time first; at fixed finite time both vanish continuously as $h \to 0$. The first is a hidden count and the second a phase, so neither is a record; their observable counterpart is the breakdown of the classical limit after the Ehrenfest time, which Theorem 3 exhibits inside the fold interval for inertial motion. The all-time limit therefore supports the thesis only through crossing streams, which is already Theorems 3–4.

6. What the thesis is, and what it needs

- **Galileo's comparison with statistical preparations has no gap.** Theorem 1 makes the record family constant in h. Any gap argument run on the parabola with Gaussian preparations and marks will fail, for the same reason the deformation families fail.
- **Determinate preparations whose streams cross have the gap, in complete records.** Theorems 2 and 3 prove it on every finite partition and exactly for inertia with graded velocity inside its fold interval; Theorem 4 proves it for a Kepler swarm whose windings overlap, in the anomaly record. Beyond Galileo the thesis holds in this topology wherever streams cross; no universal late-time statement is made.
- **Hooke and Kepler separate.** In the anomaly record, Proposition X's isochronous force keeps the zero branch for all time (Theorem 4(a)), while Kepler's force loses it for a swarm with a spread of periods once its windings overlap, at the lapping time of order $T^2/\Delta T$ for a narrow swarm spanning more than half a turn (Theorem 4(b), (c)). The gap is a property of the force law and of the swarm.
- **Single bodies are immune at fixed time.** A single packet has action spread of order h, so its lapping and fold times grow like $1/h$ (§4b); the gap concerns swarms of h-independent extent.
- **Two limits of one family.** Read as $h \to 0$ at fixed classical data with exact preparations and records, the family has no limit where coherent streams cross (Theorems 2–4). Read as the large-action regime at fixed h, with physical imprecisions held fixed, every imprecision action exceeds h and Newton is recovered (large-action proposition, §3c). Newton's mechanics is the second limit, and that limit is defined by comparison with h.
- **Smeared records and statistical preparations reach zero.** This is the content of the weak-limit theorems named in §1, and of (4).
- **Refinement with records reverses the roles.** The zero theory is the regular side of that limit.
- **The thesis is an order of limits.** By the Corollary to Theorem 2, a record of resolution δ reaches Newton if $h/(\delta\Delta p) \to 0$. "Complete records" means $\delta \to 0$ before $h \to 0$: the record limit is taken first, as the continuum limit is taken first in the mass gap.
- **The invariant discontinuous at zero is a contrast, not a spectral gap.** The leading fringe visibility $2\sqrt{\rho_1\rho_2}/(\rho_1 + \rho_2)$ of a complete record where two streams cross is fixed by Newton's branch densities, the same along the family $h > 0$, and zero at $h = 0$ (§3b). The dilation symmetry makes it dimensionless; it generates no scale. In this form the thesis asserts a discontinuity of the family at zero in the complete-record topology; a positive Newtonian action scale is not derived from it.
- **Both ends must be exact.** The leading contrast is damped by $|\hat{\varphi}(\delta\Delta p/h)|$ at the record and by $|\hat{w}_0(\Delta\,\Delta q_0/h)|$ at the preparation (§3c): it is full when both imprecision actions are small compared with h, Gaussian contrast vanishes when either ratio grows without bound, and finite ratios give partial contrast. Newton's construction has both imprecisions zero; a classical swarm attached as a mixture of localized states has thickness $\sqrt{h}$ and no Gaussian contrast, while components of any h-independent coherence length σ keep the h-independent partial contrast $e^{-\Delta q_0^2/(8\sigma^2)}$ (coherence-length proposition, §3c), and the incoherent attachment with vanishing widths converges to Newton's density at every fixed time (Theorem 5): the second premise is mutual coherence between the launch points of

crossing streams at each h, and it is necessary.

The thesis is therefore exactly as strong as two premises: that records are complete, the densities themselves rather than their smearings, and that preparations are determinate, one velocity at each place. Neither is foreign to Newton. A determinate preparation is the Newtonian default, every body with its own velocity; the statistical preparation of Theorem 1 is the extra assumption the Planck paper makes with its Gaussian marks. The second is the fifth postulate's joint determinacy (fifth-postulate note) read as a preparation, Newton's *velocitas ultima* assigned to every particle of the ensemble. The first is the records reading already required by Leibniz continuity. It is also Newton's own topology. His limits are exact: the Section I scholium asks the reader to think of the vanishing quantities "semper diminuendas sine limite", always diminishing without limit, and the geometry of the comparison has no floor at any resolution (Planck paper, §2, Proposition 1). Newton's construction contains no apparatus resolution; the ultimate ratio is the exact record, and the resolution limit is taken inside the geometry before any constant enters. Read in that topology, the $h > 0$ family is Theorem 2(i)'s complete record. The smeared topology adds a resolution δ that the construction does not contain; the Planck paper's marks are that modern addition. The gap claim and the Leibniz claim rest on the same reading of records, and the gap claim adds determinacy of the preparation. Under those two premises the statement "Newton exists only with a positive scale" has the precise form: the construction without branch data exists for every $h > 0$ and refines without extra data (1998-conjecture note, Theorem 2(b)), while at $h = 0$ it needs a branch rule (Theorem 1 there) and, wherever its streams cross, is not the limit of the $h > 0$ records (Theorems 2–4 here). With either premise dropped, the zero branch is reached. The determinate classical swarm does not by itself select the coherent attachment (5) over the mixture of localized states; that selection, an h-independent coherence length between the launch points of crossing streams (§3c), is the quantum half of the second premise and remains a premise.

7. Consequence for STATE

The gap thesis is located, refereed, and has its central-force case. It holds for complete records of determinate preparations wherever their streams cross (Theorems 2–3), and in the anomaly record it separates the force laws: Hooke keeps the zero branch for all time, Kepler loses it once a swarm's windings overlap, at the lapping time of order $T^2/\Delta T$ for a narrow swarm spanning more than half a turn (Theorem 4). It fails for Galileo's comparison with Gaussian preparations and marks (Theorem 1), for smeared records, and for refinement with records; a record of resolution δ reaches Newton if $h/(\delta\Delta p) \to 0$. The invariant discontinuous at zero is the leading fringe visibility $2\sqrt{\rho_1\rho_2}/(\rho_1 + \rho_2)$ of a complete record where two streams cross, fixed by Newton's branch densities, the same along the family $h > 0$ and zero at $h = 0$ (§3b), with full contrast when the record and preparation imprecision actions $\delta\Delta p$ and $\Delta\,\Delta q_0$ are both small compared with h (§3c); it is a contrast, not a spectral gap, and generates no scale. Consequences: the Planck paper's Gaussian parabola cannot carry the gap argument, which belongs to the Kepler swarm; the two premises, complete records as the record-limit-first order and determinate preparations as Newton's default together with an h-independent coherence length between the launch points of crossing streams, are the gap's hypotheses and the next things to defend, the first with the Leibniz records reading. The large-action proposition of §3c separates the two readings of the family: Newton's mechanics is the large-action regime of the theory with $h > 0$, defined by comparison with h, and not the $h \to 0$ limit with exact data. At $h > 0$ the Coulomb and Kepler Hamiltonians are self-adjoint and their dynamics global [@Kato1951; metadata], where the classical problem has collision singularities and the classical-first construction needs a branch rule (1998-conjecture note, Theorems 1 and 2(b)); the theory with $h > 0$ is self-sufficient where Newton's axioms leave motion undefined. Two theories must be kept apart here. Theorem 5 concerns nineteenth-century classical mechanics, the Liouville

transport of a swarm along a smooth complete flow, and shows it is reached from the $h > 0$ theory along recorded or imprecise preparations. The thesis in its strong form concerns Newton's mechanics itself, the theory of motion from his axioms, which is incomplete where the classical flow fails; that it requires $h > 0$ to exist as a theory, with the large-action regime as its only completion, is not decided by Theorem 5 and is the claim the self-sufficiency contrast begins to test. Theorem 5 proves that the gap requires the coherence premise: the incoherent attachment of the same swarm, with widths vanishing with h, reaches Newton's density at every fixed time, exactly for quadratic forces. The referee's scope verdict stands: no universal persistence after folds or lapping, no necessary resolution threshold, and no positive Newtonian action scale are proved here. Further exact cases available: the radial record of a Hooke swarm at a turning passage, and the classical-first route read with ε in place of h, where Theorem 2(i) says the unselected construction at zero exists only for single-path problems. Reject a gap argument run inside a deformation family, on Gaussian preparations of the parabola, or on smeared records, and reject presenting the non-commutation of $h \to 0$ with $t \to \infty$ as a record without an observable.